Final Outcome

Electric Vehicle Market Performance & Insights

This document summarizes the project deliverables, analytical coverage, current completion status, limitations, and future improvements. It describes the current state of the Tableau Public project accurately: the published workbook is accessible, while some shared-view formatting issues remain open for correction.

1. Project Outcome

The project transforms a prepared electric-vehicle market dataset into an interactive Tableau Public dashboard and Story. The workbook is designed to help users explore sales performance, manufacturer comparisons, regional and category patterns, geographic distribution, battery characteristics, driving range, and price variation. The Story concludes with a text-based dashboard that consolidates key insights, strategic recommendations, and the overall conclusion.

2. Final Deliverables

- Prepared combined dataset: ev_market_preprocessed_combined.csv.
- Supporting data-quality report: ev_market_data_quality_report.csv.
- Preprocessing notes: ev_market_preprocessing_notes.txt.
- Tableau Public workbook containing the main dashboard and analytical worksheet views.
- Tableau Story containing the main dashboard, nine analytical views, and the final Key Insights, Strategic Recommendations & Conclusion dashboard.
- Separate project-phase documents covering ideation, requirement analysis, planning, design, development, performance testing, final outcome, project documentation, and project demonstration.

3. Analytical Coverage

Analysis area	Purpose
Sales Trend	Explore changes and fluctuations in the sales measure across the available time field.
Top 10 Manufacturers	Compare manufacturers by the selected sales or market measure.
Region vs Category	Compare category patterns across regions.
Manufacturer Treemap	Explore relative contribution across manufacturers.
Category-wise Sale	Compare category shares of the selected sales measure.
EV Distribution Map	Explore geographic distribution where valid geographic data is available.
Capacity vs Range	Explore whether battery capacity and driving range show an association in the dataset.
Battery Tech Comp	Compare the representation of available battery technology groups.
Price Distribution	Review price variation, spread, and potential outliers.

4. Current Project Status

Published workbook	**Accessible on Tableau Public. After Tableau Public Desktop froze during development, the**

	project was published and work continued in the web version.
Dashboard and Story	The published version contains the dashboard and Story, including the final insights, strategic recommendations, and conclusion dashboard.
Presentation quality	Some KPI cards and visual formatting/layout appear distorted in the shared view. Layout correction is planned.
Performance testing	A test plan has been prepared. KPI/chart accuracy, filter behavior, and post-fix display checks still need verification.
Local Desktop recovery	Not confirmed. The accessible published workbook is the current preserved working version.

5. Key Insights and Strategic Recommendations

The final Story dashboard is intended to summarize evidence-based takeaways from the visual analysis. Specific numerical findings, manufacturer rankings, growth rates, regional leaders, and correlation strengths should be reported only after checking the exact results displayed in Tableau. This document therefore records the analytical coverage and recommendation approach without inventing numerical claims.

- Use the Sales Trend view to identify periods that merit closer review, and confirm the pattern before describing growth or decline.
- Use manufacturer rankings and the treemap to assess whether the selected measure is concentrated among a few manufacturers.
- Compare regional and category patterns before proposing region-specific marketing or distribution priorities.
- Use Capacity vs Range to explore association, not to claim that battery capacity alone causes a particular driving range.
- Use Price Distribution to review price spread and outliers, taking vehicle category and specifications into account.
- Distinguish observed data patterns from proposed business actions, and ensure each recommendation is supported by the displayed analysis.

6. Limitations

- The prepared dataset contains 299,999 rows and 38 columns, according to the preprocessing notes.
- One exact duplicate was removed during data preparation, and the record_source field was added to distinguish source records where needed.
- A total of 28,507 missing cells were retained rather than automatically imputed; missingness may affect some summaries and visualizations.
- Interpretation depends on the source field definitions and units. A distinct vehicle_id count should not be described as a count of commercial EV models unless the data dictionary confirms that meaning.
- Some KPI cards and visuals currently have shared-view formatting/layout issues.
- Local Desktop recovery has not been confirmed; the accessible published workbook is the current working reference.
- Performance tests remain partly pending, so the project should not yet be described as fully validated.

7. Future Improvements

- Correct KPI card alignment, container spacing, worksheet fit settings, and dashboard sizing for the intended sharing view.
- Reopen the published link after layout changes to confirm the updated presentation.

- Complete verification of KPI calculations, chart aggregations, filters, and Story navigation.
- Improve tooltips, labels, captions, and units wherever they help viewers interpret the charts.
- Review missing-value treatment and document any exclusions or assumptions used in analysis.
- Add verified numerical findings and concise, data-supported recommendations to the final Story dashboard.
- Keep an independent downloadable backup of the workbook where supported by the Tableau Public workflow.

8. Conclusion

The project provides a Tableau-based exploration of EV market performance and vehicle characteristics, supported by a Story that brings the visual analysis together in a final insights and recommendations dashboard. The published workbook remains accessible and preserves the current project progress. Before the project is considered fully presentation-ready, the known shared-view formatting issues should be corrected and the remaining performance checks completed. Once those tasks are verified, the project can serve as a portfolio example of data preparation, visual analytics, and evidence-based business interpretation.

Current status: Published and accessible; layout refinement and remaining validation are in progress.